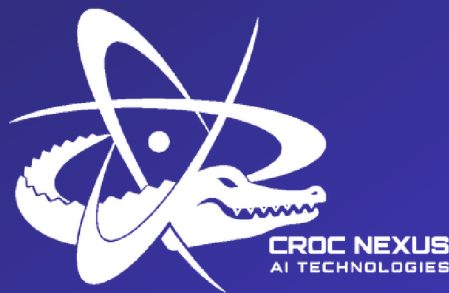


MAIC NEXUS CHALLENGE 2026



Croc Sentinel

AI-Powered Emergency Response & Smart Monitoring
Platform for Public Facilities

Malaysia AI Innovation Platform

Track: AI Public Services & Smart City

Submission Document · June 2026 · Croc Nexus AI Technologies

1 Executive Summary

Croc Sentinel is a production-ready, mobile-first IoT security platform developed by **Croc Nexus AI Technologies** for Malaysia's public infrastructure and smart city ecosystem.

The platform connects panic buttons, motion sensors, sirens, and CCTV triggers to a unified cloud backend — delivering a real-time command centre on iOS and Android. Operators see every device, every alarm, and every site on a single Interactive GIS dashboard, with push notifications arriving within 3–30 seconds depending on signal strength.

Unlike traditional siloed alarm systems designed for single commercial buildings, Croc Sentinel enables **multi-site, multi-agency monitoring** across schools, hospitals, government buildings, and municipal CCTV networks. Municipal councils, school boards, and hospital administrators can share device groups across agencies while maintaining role-based access control.

3–30s

Alert Delivery (Signal-Adaptive)

24/7

Continuous Monitoring

Multi

Site GIS Dashboard

iOS+

Android Native App

Target Public Facilities

Croc Sentinel is designed for facilities where public safety is non-negotiable and response time directly affects outcomes:

City CCTV Networks

Schools & Campuses

Hospitals & Clinics

Government Buildings

Community Centres

Municipal Sites

DBKL Operations

Public Transport Hubs

MAIC Nexus 2026 Alignment

- **Track:** AI Public Services & Smart City — intelligent monitoring and emergency response for public facilities
- **Status:** Production cloud API, live IoT edge hardware, and mobile app already operational — not a concept prototype
- **Scale:** Deployable from a single campus to a city-wide municipal facility network with dozens of sites
- **Impact:** Replaces 3–5 fragmented monitoring tools with one affordable, mobile-first platform

For Municipal Agencies

Centralised visibility across all public sites. Cross-agency group sharing lets DBKL, school boards, and hospital admin coordinate without replacing existing hardware.

For Facility Operators

One mobile app for daily monitoring and emergency response. ARM ALL / DISARM ALL, live event timeline, GPS patrol tracking, and instant push alerts.

Key Takeaway

- Live, production-ready platform — real hardware, real API, real mobile app deployed today
- Unifies fragmented facility security into one mobile command centre with 3–30 second signal-adaptive alerts
- Built specifically for schools, hospitals, government sites, and city CCTV networks across Malaysia

2 Problem Statement

Public facilities in Malaysia face a shared crisis: **fragmented security systems that fail to communicate in real time.**

When a panic button is pressed at 11 PM in a school corridor, minutes are lost because alarms do not reach the right people quickly enough. CCTV operators watch one screen, guard patrols carry another radio, and building alarms feed a third panel — none of which talk to each other.

Existing enterprise solutions carry prohibitive per-site pricing, are designed for single commercial buildings, and lack unified map-based views across multiple municipal locations. Public schools and community centres simply cannot afford them.

Key Pain Points by Facility Type

Schools & Campuses

Panic buttons may only ring a local buzzer. Principals and security guards are not notified simultaneously. No campus-wide siren activation from a mobile device. After-hours intrusion goes undetected for 15+ minutes.

Hospitals & Healthcare

Emergency alarms in restricted zones are isolated from municipal dispatch teams. No cross-agency visibility when incidents require police or fire coordination. Manual phone trees delay critical response.

Government Buildings

Multi-site council offices each run independent alarm systems. No affordable centralised monitoring dashboard. Security staff cannot see alarm status across all branch offices from one app.

City CCTV Networks

Camera motion alerts are not correlated with on-site IoT sensors and sirens. Operators must switch between CCTV software and separate alarm panels. No unified GIS view of all triggers city-wide.

Root Causes

- **Siloed Systems:** CCTV, alarms, and sensors purchased from different vendors with no integration layer
- **Phone-Tree Delays:** Human relay chains add 5–15 minutes before the right responder is notified
- **No Mobile Command:** Security staff cannot arm, disarm, or respond to alarms from their phone in the field
- **High Cost Barrier:** Enterprise solutions exclude public schools and small municipal sites from smart security
- **No Audit Trail:** Incidents lack timestamped event logs required for compliance and post-incident review

Critical Scenario — School Intrusion at 11 PM

A motion sensor triggers in a school corridor. The guard's phone never receives the alert because the sensor and notification system are on separate platforms. By the time the guard arrives, 12 minutes have passed. Croc Sentinel exists to eliminate this gap — delivering the alert to every authorised operator within 3–30 seconds depending on signal strength.

Key Takeaway

- Alarms, CCTV, and sensors operate in isolation — no unified command layer exists today
- Seconds matter in emergencies — delayed response costs lives, property, and public trust
- Municipal agencies need one affordable, scalable, mobile-first platform built for public facilities

3 Why Choose Croc Sentinel

Traditional security systems were built for single commercial buildings with dedicated control rooms. Croc Sentinel is purpose-built for **public facilities and smart city operations** — faster alert delivery, accessible deployment, and AI-powered event intelligence in one unified mobile platform.

We are the only solution in this competition space combining production IoT deployment, Interactive GIS multi-site command, and AI visual threat assessment — all accessible from an iOS and Android app that operators already carry in their pocket.

Traditional Systems

- ✗ 5–15 min alert delay via phone tree
- ✗ Single building only — no multi-site map
- ✗ Prohibitive per-site setup cost
- ✗ Separate CCTV, alarm & sensor tools
- ✗ No AI event classification
- ✗ Vendor lock-in, long deployment cycles

VS

Croc Sentinel

- ✓ 3–30 second push to all operators (signal-adaptive)
- ✓ Multi-site Interactive GIS dashboard
- ✓ Affordable per-site IoT edge deployment
- ✓ One unified mobile app for everything
- ✓ AI camera scoring & auto escalation
- ✓ QR activation — deploy in minutes

Capability	Traditional	Croc Sentinel
Alert Delivery Speed	5–15 minutes (phone tree)	3–30 seconds (signal-adaptive, live + push)
Multi-Site GIS Map	Not available	Live GIS dashboard with alarm pins
Mobile Command App	Limited / desktop only	Full iOS & Android native app
AI Event Intelligence	Manual operator monitoring	Camera AI scoring & auto call escalation
Deployment Accessibility	Enterprise-only pricing	Public-sector affordable deployment
Production Status	Months to deploy	Live today — QR activation

Instant Alerts

Dual-channel delivery ensures every operator receives push notifications within 3–30 seconds depending on signal strength

GIS Command

Multi-site map with live alarm pins, 3D site zoom, and per-site device counts

Affordable

Low-cost IoT edge hardware with significantly lower total deployment cost than traditional enterprise systems

AI-Powered

Camera capture, danger scoring, nearest-responder dispatch, and auto phone call

Public-Ready

Designed for schools, hospitals, government, and municipal CCTV networks

Production Live

Real IoT hardware, live cloud API, and App Store / Play Store ready mobile app

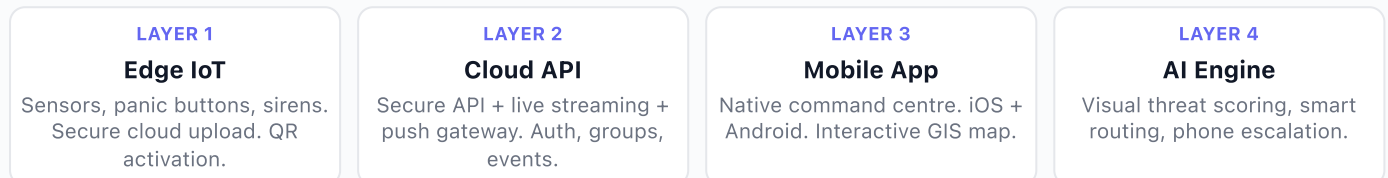
Why Us — At a Glance

- 10x faster alert delivery than traditional phone-tree systems
- Significantly lower deployment cost with affordable IoT edge hardware
- Only platform combining IoT + GIS + AI + mobile command for public facilities

4 Our Solution & Architecture

Croc Sentinel delivers an end-to-end stack: affordable IoT edge hardware, a secure production cloud API, and a native mobile command app — unified under one platform with affordable public-sector deployment.

Each layer is independently scalable. Add new edge devices via QR activation without touching the cloud. Add new operator accounts without redeploying hardware. The mobile app supports over-the-air updates for rapid feature delivery.



Three Integrated Layers

- **Edge:** IoT edge controllers with wireless connectivity — panic buttons, motion sensors, sirens, relay modules. QR activation in minutes.
- **Cloud:** Secure API endpoints, live data streaming, push notification gateway, HTTPS encryption, role-based access (Super Admin / Site Admin / Operator), cross-agency group sharing.
- **Mobile:** Overview dashboard with breathing-border alarm indicators, GIS map with 3D site zoom, filterable event timeline, device management, GPS patrol, ARM ALL / DISARM ALL.

Data Flow & Reliability

- **Upstream:** Edge Device → Cloud Hub → AI Classification → Live Broadcast + Push Alerts to all operators
- **Downstream:** Mobile App → Cloud API → Device Command → Edge Devices (arm / disarm / siren activation)
- **Dual-Channel:** Live streaming and push alert redundancy ensures delivery even if one channel fails
- **Offline Detection:** Heartbeat monitoring flags device connectivity loss before an incident occurs

Technology Stack

Cross-platform mobile framework, efficient state management, interactive GIS engine, background location services, secure credential storage.

Security

HTTPS-only API, secure session cookies, role-based access control, encrypted token storage, multi-tenant site isolation.

IoT Edge

Cloud Messaging

Mobile App

Live Streaming

Push Alerts

GIS Mapping

Secure API

AI Engine

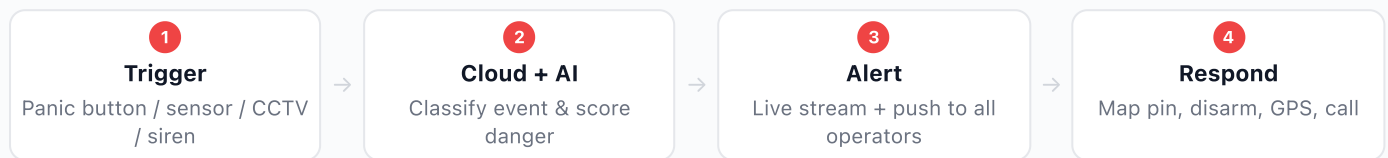
Key Takeaway

- Four independent layers — edge, cloud, mobile, AI — each scales horizontally across city-wide deployments
- Full platform at significantly lower cost than traditional enterprise security systems

5 Emergency Flow & Use Cases

A four-stage automated pipeline compresses emergency response from minutes to seconds — with full audit trail at every step.

From the moment a sensor triggers to the moment every authorised operator receives a push notification and sees the alarm pin on the map, the entire cycle completes within 3–30 seconds depending on signal strength. No phone tree. No manual relay. No delay.



Stage-by-Stage Detail

Stage 1 — Trigger

Edge devices detect panic button press, motion, or siren activation. Secure message sent to cloud in milliseconds. Device heartbeat confirms online status.

Stage 2 — Cloud + AI

Backend classifies event type and AI scores danger from camera capture. Routes to correct notification group based on zone and agency rules.

Stage 3 — Alert Delivery

Dual-channel notification: live stream updates all open app sessions; mobile push delivers to background sessions with deep linking.

Stage 4 — Response

Operator sees alarm pin on GIS map, reviews timeline, triggers ARM ALL for sirens, shares GPS for patrol dispatch. Full audit log recorded.

Public Facility Use Cases

Schools & Campuses

Panic buttons in corridors and staff rooms. Mobile push notifies principals and guards simultaneously. ARM ALL activates campus-wide sirens from any authorised phone.

Hospitals

Monitor restricted zones and emergency stations. Role-based access separates hospital security from municipal dispatch. Cross-agency sharing enables police coordination.

City CCTV

Integrate CCTV motion triggers with on-site IoT sensors and sirens. Municipal operators view all camera-linked devices on a live GIS dashboard city-wide.

Government & Community

Multi-site dashboard for council offices and community centres. QR activation allows rapid rollout. Event timeline provides audit logs for compliance.

- **ARM ALL:** Any authorised operator activates campus-wide sirens from the Overview dashboard; DISARM ALL resets when resolved
- **Deployment:** Single campus to city-wide network — QR-code activation in minutes, no technician required

Key Takeaway

- Trigger → Cloud → Alert → Respond completes within 3–30 seconds (signal-adaptive) with full audit trail
- Schools, hospitals, CCTV, government, and community facilities — one unified platform

6 AI & Intelligent Features

Intelligence is embedded at the event-processing layer — not bolted on as an afterthought. When an alarm triggers, Croc Sentinel captures camera footage and sends it to AI for autonomous danger assessment, escalation, and follow-up.

The AI engine analyses every device event and linked camera capture before it reaches operator screens. The system assigns a real-time danger score, decides whether human intervention is required, and escalates automatically when alarms are not acknowledged in time.

AI Visual Threat Assessment

Camera Capture → AI Analysis

On alarm trigger, live camera footage is captured and sent to the AI engine. The system classifies the scene and calculates a danger coefficient from low to critical based on visual evidence and event context.

Autonomous Decision

AI independently determines false alarm, low-priority alert, or genuine threat. Only events exceeding the danger threshold proceed to operator escalation — reducing alert fatigue.

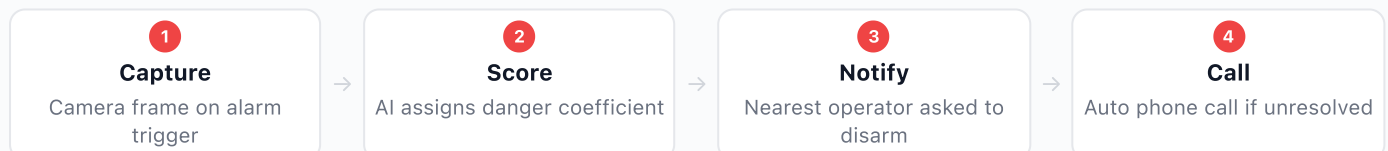
Nearest-Responder Dispatch

When danger is high, AI contacts the **nearest available operator** via push alert, requesting them to attend the site and **disarm the alarm** through the mobile app.

Phone Call Escalation

If the alarm remains active after a configurable reminder window and has not been disarmed, the system **automatically initiates a phone call** to the designated on-duty contact.

Autonomous Escalation Flow



Deployed Intelligence & Pipeline

- **Classification:** Parses panic, siren, motion, and offline events — filters false alarms from environmental triggers
- **Smart Routing:** Maps devices to notification groups by zone, agency, and role — including nearest-responder selection
- **Capture → Analyse → Escalate → Follow-up → Log:** Full pipeline from camera snapshot to phone call, every step timestamped for audit
- **Future Roadmap:** Anomaly detection on event timelines, automated compliance reports, predictive device health scoring

Key Takeaway

- AI captures camera footage, scores danger autonomously, escalates to nearest operator — then calls if unresolved
- Full audit trail from capture to phone escalation — supporting smart public safety and municipal digital service goals

7 Smart City Impact & Team

Expected Impact on Public Safety

10×

Faster Alert vs Phone Tree

Low

Public-Sector Cost Profile

1 App

Replaces 3–5 Tools

100+

Sites on One Map

- **Public Safety:** 3–30 second signal-adaptive panic alarm propagation ensures every authorised responder is notified simultaneously
- **Multi-Agency Coordination:** Role-based access with cross-agency sharing lets DBKL, school boards, and hospital admin coordinate without replacing existing infrastructure
- **Operational Efficiency:** One mobile app replaces separate CCTV viewer, alarm panel, sensor dashboard, and manual phone tree
- **Digital Inclusion:** Affordable IoT edge hardware makes smart security accessible to public schools and community centres previously excluded by enterprise pricing
- **Compliance & Audit:** Timestamped event timeline provides the audit trail municipal agencies need for incident reporting and regulatory compliance

Development Roadmap

Phase 1 · Now

Core platform production-deployed — IoT edge devices, Interactive GIS dashboard, dual-channel alerts, iOS/Android mobile app, AI visual threat assessment, QR activation, ARM ALL / DISARM ALL, GPS patrol.

Phase 2 · 2026

AI anomaly detection on event timelines. CCTV correlation with on-site sensor data. Municipal analytics dashboard with incident heatmaps and response time metrics.

Phase 3 · 2027+

Open integration API for emergency dispatch systems. Integration with Malaysia emergency services. Nationwide rollout partnership with state governments and municipal councils.

Team & Contact — Croc Nexus AI Technologies

Our team spans IoT edge hardware, full-stack cloud engineering, mobile application design, and AI event processing — building production systems, not prototypes.

Team Name	CROC Nexus AI Technology
Contact Email	partnerships@crocnexus.com
Contact Phone	+084-349525

IoT Engineering

Edge firmware, sensor integration, QR activation

Cloud Development

Live streaming, push gateway, multi-tenant auth

Mobile App Design

Interactive GIS, push notifications, responsive UI

AI Integration

Visual threat scoring, smart routing, auto call

Public Safety

Schools, hospitals, government, CCTV expertise

Smart City

Multi-agency coordination and municipal scalability

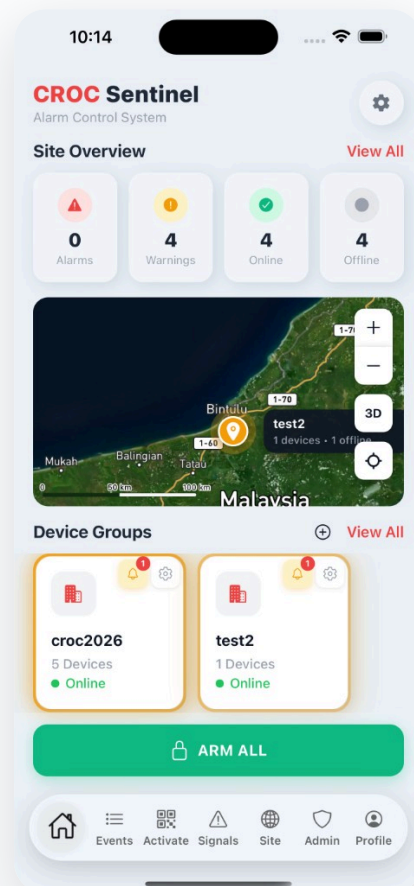
Conclusion

- Croc Sentinel — the unified, AI-ready smart city platform built for Malaysia's public facilities
- Protecting schools, hospitals, government buildings, and CCTV networks when every moment counts
- We invite judges and municipal partners to explore a live demonstration of the production platform

8 App Demo — Overview Dashboard

Production app screenshot — the Overview screen is the mobile command centre showing all device groups, live alarm status, embedded GIS map, and one-tap ARM ALL / DISARM ALL controls.

Site Overview cards display alarms, warnings, online and offline device counts at a glance. Group cards show device counts with breathing-border indicators — red for armed/alarm state, green for safe. The embedded map provides multi-site pins with live device status for city-wide situational awareness.



Site Overview

Alarms, warnings, online/offline counts updated in real time via live streaming

Embedded GIS Map

Multi-site map with device pins, 3D zoom, and per-site status badges

Device Groups

Group cards with live device counts, notification badges, and zone settings

ARM ALL Control

One-tap campus-wide siren activation from the bottom action bar

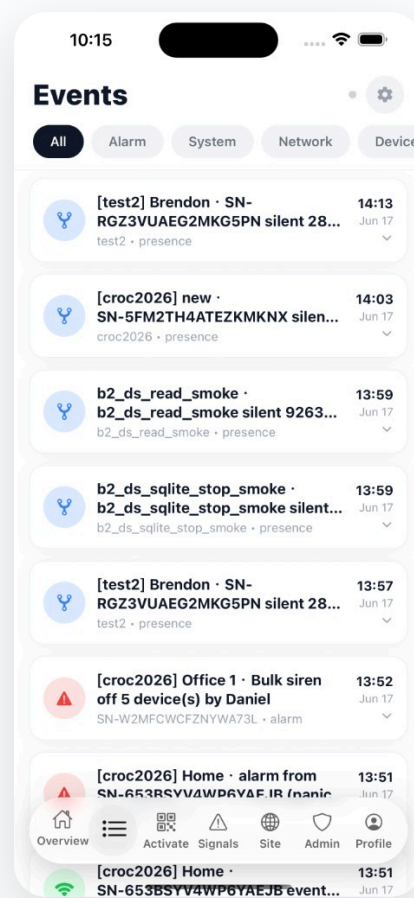
Demo Highlight

- Real production app — not a mockup. Overview replaces the traditional desktop control room on mobile.

9 App Demo — Event Timeline

Production app screenshot — chronological event log with live streaming for every panic button press, motion trigger, siren activation, and system event across all sites.

Filter by Alarm, System, Network, or Device for fast incident review. Each entry shows group name, location, device serial number, event type, and precise timestamp — providing the full audit trail municipal agencies require for compliance and post-incident reporting.



Live Event Stream

Events appear instantly without manual refresh — powered by live data streaming

Smart Filters

Filter by alarm, system, network, or device type for targeted review

Alarm Highlighting

Panic button and alarm events highlighted in red with location and serial number

Deep Navigation

Tap any event to navigate directly to the source device and zone details

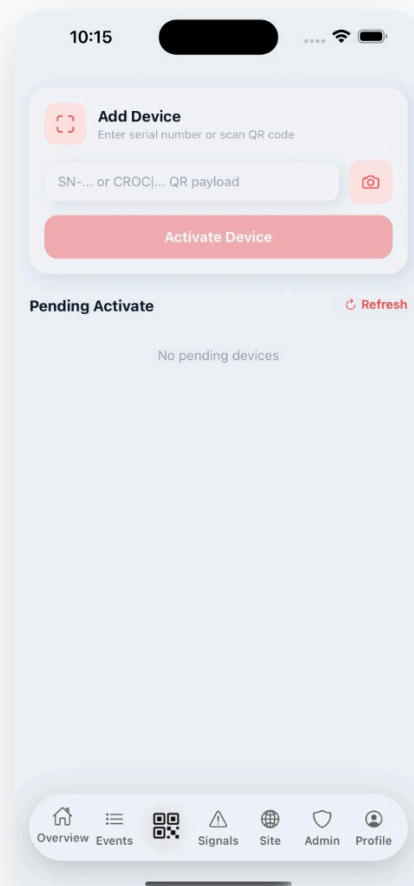
Demo Highlight

- Every trigger, alert, arm, and disarm action is timestamped — full audit trail for municipal compliance.

10 App Demo — Device Activation

Production app screenshot — onboard new IoT edge devices by entering a serial number or scanning a QR code, with no technician visit or complex configuration required.

The Add Device screen accepts SN- prefixed serial numbers or CROC QR payloads. The Pending Activate queue lets Site Admins review and approve new devices before they join a site group — ensuring only authorised hardware enters the network.



QR Code Scan

Camera scan button for instant device onboarding in the field

Serial Entry

Manual SN- or CROC| payload entry for backup activation

Pending Queue

Review and approve devices before they go live on the network

Rapid Deployment

Non-technical staff can activate devices in minutes — no IT visit required

Demo Highlight

- QR activation enables rapid rollout across public schools and community centres without specialist technicians.

11 App Demo — Signals & Alarm Routing

Production app screenshot — the Signals dashboard provides activity monitoring, alarm fan-out routing analytics, and 24-hour / 7-day alarm statistics for operational oversight.

Panic button and alarm fan-out events are highlighted with location tags (e.g., Office2, Mafrica1), device serial numbers, and fan-out routing status. Administrators can identify top alarm sources and response patterns across the entire municipal network from one screen.



Alarm Analytics

24h and 7d alarm counts with top source device tracking

Panic Button Events

alarm_fanout events highlighted with location and serial number tags

Routing Visibility

Fan-out count and email routing status per signal event

Bulk Operations

bulk_siren_off and group_card_apply events logged with operator name

Demo Highlight

- Signals screen connects AI escalation logic to real alarm routing — the operational backbone of Croc Sentinel.
- Production platform ready for live demonstration — contact Croc Nexus AI Technologies.